

Coordinate Subspaces of the HRST Norming-M-Basis Space

Run `fa_banach_001`, agent `agent_lane_12`

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Source Problem

Hajek and Russo ask whether admitting a norming Markushevich basis is hereditary to subspaces [1, Problem 6.1]. The problem appears on PDF page 24 of arXiv:2402.18442 and in the parsed source at `manuscript.tex:685-691`.

The same source discusses the Hajek–Russo–Somaglia–Todorčević construction: there is an Asplund space X_ϱ with a 1-norming M-basis which is not WCG [1, Theorem A, recalled as Theorem 4.4]. This makes X_ϱ a natural place to search for a possible bad subspace in Problem 6.1.

Result

The coordinate-generated subspaces of X_ϱ are not counterexamples.

Theorem 1. *Let $X_\varrho = \overline{\text{span}}\{f_\alpha : \alpha < \omega_1\} \subset C(K_\varrho)$ be the HRST space as recalled in [1, Theorem 4.4]. For every $A \subseteq \omega_1$, put*

$$X_A = \overline{\text{span}}\{f_\alpha : \alpha \in A\} \subseteq X_\varrho.$$

Then

$$\{f_\alpha; \mu_\alpha|_{X_A}\}_{\alpha \in A}$$

is a 1-norming M-basis of X_A .

Proof Intuition

The HRST construction is unusually rigid on its coordinate subspaces. The compact K_ϱ is the closure of a family $\mathcal{F}_\varrho$ of finite subsets of ω_1 . On X_ϱ , evaluation at any finite set $F \in \mathcal{F}_\varrho$ is exactly the finite sum of the coordinate measures indexed by F . If we keep only coordinates in A , the same evaluation is simply the finite sum over $F \cap A$. Since the finite sets $\mathcal{F}_\varrho$ are dense in K_ϱ , those evaluations still norm the restricted space.

Proof

Proof. Recall the notation from the source paper. The compact space K_ϱ is the pointwise closure of

$$\mathcal{F}_\varrho = \{F_n(\alpha) : n < \omega, \alpha < \omega_1\} \subseteq [\omega_1]^{<\omega}.$$

For $\alpha < \omega_1$, the coordinate function $f_\alpha \in C(K_\varrho)$ is defined by

$$f_\alpha(B) = \begin{cases} 1, & \alpha \in B, \\ 0, & \alpha \notin B, \end{cases} \quad B \in K_\varrho,$$

and $\mu_\alpha = \delta_{\{\alpha\}}$ is the point mass at the singleton $\{\alpha\} \in K_\varrho$. The system $\{f_\alpha; \mu_\alpha|_{X_\varrho}\}_{\alpha < \omega_1}$ is the canonical 1-norming M-basis of X_ϱ in the HRST theorem.

Fix $A \subseteq \omega_1$ and define X_A as in the statement. The displayed system is biorthogonal because, for $\alpha, \beta \in A$,

$$\mu_\alpha(f_\beta) = \delta_{\{\alpha\}}(f_\beta) = f_\beta(\{\alpha\}) = \delta_{\alpha\beta}.$$

It is fundamental by the definition of X_A .

It remains to prove the 1-norming property. Let $F \in \mathcal{F}_\varrho$. Since F is finite, the evaluation functional $\delta_F|_{X_A}$ satisfies

$$\delta_F|_{X_A} = \sum_{\alpha \in F \cap A} \mu_\alpha|_{X_A}.$$

Indeed, both sides agree on every generator f_β with $\beta \in A$: they both give 1 if $\beta \in F$ and 0 otherwise. Hence the equality holds on the dense linear span of the generators and therefore on X_A . Moreover, $\|\delta_F|_{X_A}\| \leq 1$.

For $x \in X_A$, continuity on the compact space K_ϱ and density of $\mathcal{F}_\varrho$ in K_ϱ give

$$\|x\|_{C(K_\varrho)} = \sup_{B \in K_\varrho} |x(B)| = \sup_{F \in \mathcal{F}_\varrho} |\delta_F(x)|.$$

By the preceding identity, each $\delta_F|_{X_A}$ belongs to $\text{span}\{\mu_\alpha|_{X_A} : \alpha \in A\}$ and has norm at most one. Therefore

$$\|x\| \leq \sup \left\{ |\nu(x)| : \nu \in \text{span}\{\mu_\alpha|_{X_A} : \alpha \in A\}, \|\nu\| \leq 1 \right\}.$$

The reverse inequality is immediate from the condition $\|\nu\| \leq 1$. Thus the span of the restricted coordinate measures is 1-norming for X_A . In particular it is total, and the biorthogonal fundamental system is a 1-norming M-basis. $\square$

Limitations and Novelty Check

This theorem does not settle the full heredity problem. It only treats subspaces of X_ϱ that are generated by subsets of the canonical HRST coordinate family. A bad subspace of X_ϱ , if one exists, must therefore mix the coordinates in an essentially non-coordinate way.

The result was obtained by re-reading the HRST norming argument as presented in the source survey. It appears to be an immediate but useful consequence of the finite-set evaluation formula in that construction. No claim is made that this observation is absent from the original HRST paper; its value here is as a recorded search filter for Problem 6.1.

Human Review Recommendation

Review as a positive coordinate-subspace subcase for Problem 6.1 and as a counterexample-search obstruction inside the HRST example. The proof is short; the main points are the equality $\delta_F|_{X_A} = \sum_{\alpha \in F \cap A} \mu_\alpha|_{X_A}$ and the use of density of $\mathcal{F}_\varrho$ in K_ϱ .

References

- [1] P. Hajek and T. Russo, *Norming Markushevich bases: recent results and open problems*, arXiv:2402.18442.